

MUSIC PERFORMANCE & STREAMING ANALYTICS

USING MS EXCEL AND POWER BI

1. PROJECT OVERVIEW

The music industry generates massive amounts of streaming data across digital platforms such as Spotify, YouTube, TikTok, Apple Music, Pandora, Deezer, Amazon Music, and Sound Cloud. This project transforms raw music streaming data into meaningful business insights through data cleaning, exploratory data analysis (EDA), and interactive dashboard development using Microsoft Excel and Power BI.

2. OBJECTIVES

- Clean and standardize the raw music dataset.
- Handle missing values and duplicate records.
- Validate track, album, and artist metadata using ISRC where possible.
- Analyze artist performance, song performance, platform engagement, trends, and relationships.
- Build an interactive Power BI dashboard.
- Generate business insights and recommendations.

3. DATA SOURCE

Dataset: Music Performance & Streaming Analytics

Domain: Music Streaming Analytics

Dataset Size: Approximately 4,600 songs (final cleaned dataset: 4,598 songs), 27 columns.

Platforms: Spotify, YouTube, TikTok, Apple Music, Deezer, Pandora, Amazon Music, SoundCloud, and Shazam.

4. PROBLEM STATEMENT

The music industry releases thousands of songs across multiple streaming services. This project identifies top-performing artists, songs, albums, platform contributions, release trends, and relationships between streaming metrics to support data-driven decision-making

5. DATASET UNDERSTANDING

Column Name	Data Type
Track	Text
Album Name	Text
Artist	Text
Release Date	Date
ISRC	Text
All Time Rank	Whole Number
Track Score	Decimal Number
Spotify Streams, Spotify Playlist Count, Spotify Playlist Reach, Spotify Popularity	Whole Number
YouTube Views, YouTube Likes, YouTube Playlist Reach	Whole Number
TikTok Posts, TikTok Likes, TikTok Views	Whole Number
Deezer Playlist Count, Deezer Playlist Reach	Whole Number
Pandora Streams, Pandora Track Stations	Whole Number
Apple Music Playlist Count, Air Play Spins, SiriusXM Spins, Amazon Playlist, Soundcloud Streams, Shazam Counts	Whole Number

6. DATA CLEANING & PRE-PROCESSING

- **Removed duplicate records**

[illegible]

- **Handled missing values**

Album Name	Artist	Release D	SRC	All Time R	Track Score	Spotify	Spotify	Spotify	Spotify	Pc	YouTube	YouTube	TikTok	Pc	TikTok	Lik	TikTok	Vi	YouTube	Apple	Mu	AirPlay	Sp	SinusM	Deezer	Pc	Deezer	Pc	Amazon	P	Pandora	S	Pandora	T	Soundcloud	Shaz		
Million Dollar Baby - Single	Tommy Richman	#####	QM24524K	1	72.4	39.04,09	30,716	19,631.5	92	8,42	74,17,13,126	57,67	65,16,95	93,33,82,11	15,05,97,0	210	40,975	604	62	175,98,71	114	1,80,04,65	22,931	48,14,57	26,69													
Nike Like	Kendrick Lamar	5/4/2024	USUG124K	2	54.9	31,27,83.8	28,113	17,45,97,1	92	11,63,49,0	34,66,739	6,74,700	3,52,23,54	20,83,39	10,15,83,80	3	188	40,778	3	67	104,22,43	111	77,80,028	28,444	66,27,015	11,818												
I like the way you kiss me	Aterkiss	#####	QZ8240K	3	53.8	60,110.9	22,331	21,16,07,6	92	12,25,99,1	12,28,730	35,25,400	27,51,54	23,36,91	20,37,37,84,9	190	3,83,338	536	136	3,62,21,84	172	50,22,621	5,639	72,08,651	52,185													
Flowers - Single	Miley Cyrus	#####	US3M122K	4	44.9	2,03,12,80	2,69,802	13,65,69,0	85	1,09,61,00	1,06,29,718,89,811	1,07,87,57	14,60,37,23	3,35,11,88	394	174,799	2,182	264	2,46,84,24	210	19,82,60,2	2,03,384			1,18,													
Houdini	Eminem	#####	US4G124K	5	42.3	10,70,34,9	7,223	15,49,18,8	88	73,79,35	36,70,188	16	400		112,67,83	182	12,185	1	82	17,60,60,2	105	49,48,884	7,006	2,07,179	45,17													
Love On Me	Jack Harlow	#####	USAT2221	6	40.1	67,90,58,7	105,892	17,34,21,0	83	13,11,40,1	13,92,593	42,02,367	21,49,43	42,93,86,86	2,86,72,22	138	5,22,042	4,654	86	17,67,25	152	13,85,29,3	57,982	94,98,081	45,17													
Beautiful Things	Benson Boone	#####	USWB123K	7	40.2	90,01,58,7	73,118	20,15,85,7	86	30,87,23	1,41,120,760		2,95,04,94	53,49,15,3	4,15,0,79	280	3,83,478	429	168	4,81,95,87	154	6,84,65,43	50,372	94,98,081	45,17													
Gata Only	FloayMenor	2/2/2024	QZL38240K	8	37.8	67,30,79,1	40,094	22,11,23,36,9	92	22,83,82	5,14,39,495	35,000,000	33,85,46,6	3,80,45,81	21,12,5,81	160	17,221	30	87	3,32,45,59	53	33,72,428	5,762			60,69												
Don Omner	MUSIC LAB JPN	6/9/2024	TCJA246K	9	35.7	1,65,30,18	1	15																														
BANDABAND (feat. Lil Baby)	Central Cee	#####	US3M124K	10	33.6	9,06,57,76	10,040	18,40,19,6,1	86	3,27,35,24	9,48,88,682	3,25,800	12,15,75	97,46,56	17,47,06,8	191	3,823	117	78	1,08,00,09	92	10,05,626	842	36,79,709	6,66,													
I Had Some Help	Post Malone	#####	USM724K	11	32.2	15,20,80,84	16,219	16,87,66,1	91	3,22,29,06	5,50,191	1,84,500	21,00,000	32,71,30,7	157	62,312	71	60	1,20,09,07	114	2,31,55,47	21,172	15,94,605	9,63,														
The Door	Teddy Swims	#####	USWB124K	12	32.0	21,11,55,31	2	2,63,767																														
HIT ME HARD AND SOFT	Billie Eilish	#####	USM724K	13	31.6	23,16,36,1	13,800	19,72,80,6	92	4,00,22,52	13,07,290	1,95,300	4,58,89,00	36,00,17,0	34,21,07,1	244	41,344	45	138	3,82,43,63	163	13,54,692	1,219	13,13,357	4,50,													
WE DON'T TRUST YOU	Future	#####	US3M124K	14	30.8	25,351,86	43,025	10,88,66,16	88	9,80,81	49,14,56,177		17,44,91,14							3	55	30,01,826	199	2,62,52,26	18,364	1,20,38,03	19,639											
bathroom floor	Kids With Buns	#####	NLC24210	15	301.6	2,97,185	177	24,445	12	6,73,67,65	2,16,87,85	48	3	1,088	22,294					1	1,057																	

All Time	Track Score	Spatial Stream	Spatially Partial Count	Spatially Partial Count	Spatially Partial Count	Spatially Partial Count	YouTube Views	YouTube Likes	TikTok Plays	TikTok Likes	TikTok Views	YouTube Playlist Count	Apple Music Playlist Count	AirPlay Plays	Shazam Spins	Deezer Playlist Count
1	725.4	390470996	30716	196651588	91	94214754	1713126	5787700	651565900	5332281996	150597040	210	40975	684	62	67
2	545.9	312703884	28113	174597137	92	11637404	9486739	674700	95223547	208390925	156380351	188	108	40778	3	67
3	538.6	601305283	54381	211607669	92	12259916	2228730	3025400	275154237	3969120650	937784955	190	74333	536	136	194
4	494.6	2051280633	136569078	335696078	85	1006100899	16029196	7189811	1078757968	14602725994	3381188582	994	1474799	2182	264	106
5	423.3	170340922	7223	151468974	88	73773957	3670188	16400	26571692	266064027	112768531	182	12185	1	82	182
6	410.1	670665438	10882	175421034	83	131148091	1392593	4021367	214943489	293866663	2867212632	150	522042	4654	86	106
7	407.2	900158751	73118	210585714	86	380723145	4120760	182529	29589480	54915313	4601579812	280	383478	429	168	168
8	375.8	675079153	40048	211236940	92	228832568	1439495	3500000	38354668	3804584163	2112581620	160	17221	30	87	87
9	355.7	1653018119	1	15	63	148269610	1257935	182529	26571692	266064027	97746040	0	5954	86	0	0
10	330.6	90676573	10400	184199419	86	32735244	98862	325800	112574500	974562300	177406874	191	3823	117	78	78
11	322.1	250080413	16219	1868166158	91	32219063	550191	182529	184500	2100000	327130768	157	63212	71	60	60
12	320.2	111155129	2	165767	63	148269610	1257935	182529	26571692	266064027	97746040	0	5954	86	0	0
13	316.3	221636195	13800	197280692	94	40212534	1307290	195300	45888000	360017000	342107131	244	41944	455	138	138
14	308.2	355186436	43025	108166634	88	98081499	1456177	182529	26571692	266064027	1744891415	153	90137	3	55	55
15	301.6	297185	277	24445	12	6738502312	16785489	3	1088	22234	97746040	1	1057	86	0	0
16	299.9	925655369	103605	79944821	79	934201201	4294166	11054500	521775116	7499214052	2042145659	94	91231	228	60	60
17	297.6	394834040	12784	177932568	89	104673836	2484083	139300	25348800	207317000	268893710	221	129968	3	99	99
18	293.7	1258636694	105218	186845580	87	1141646818	1514837	2384429	419319161	5456156211	2667285437	207	667329	3310	188	188
19	292.6	91272461	6499	52887545	86	16308055	569377	941900	119740900	1225345800	8839168	76	181	1	24	24
20	292.0	3301814535	407119	169738402	87	80089915	9101589	275903	163309701	15405881843	7289707052	455	1778811	4909	330	330
21	290.5	1063398331	170880	217112197	78	240270670	2746668	3081732	652432878	7095564106	1063591802	148	408999	273	143	143

- **Corrected metadata using ISRC where available**

	G	II	L
us	Artist		ISRC
			QZNWQ2410638
			QZYFZ2445017
			QZNWT2471497
			QZNWU2402635
			MX2832415361

7. TOOLS & TECHNOLOGIES

Excel: Data Cleaning and Transformation.

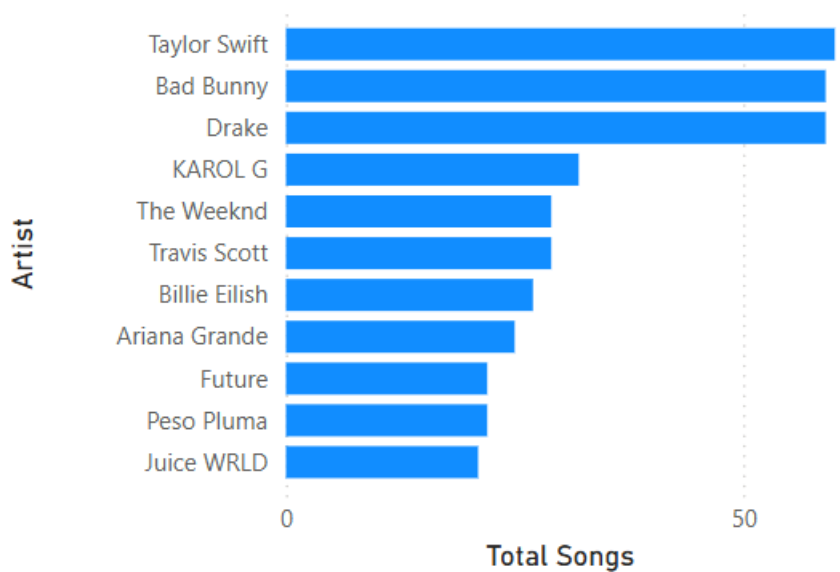
Power BI: Data Modelling, DAX Calculations, Visualization, And Interactive Dashboard Creation.

8. EXPLORATORY DATA ANALYSIS (EDA)

EDA included:

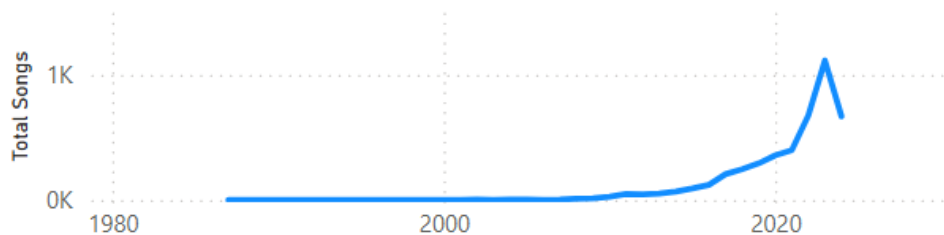
- Distribution of artists and song

Total Songs by Artist



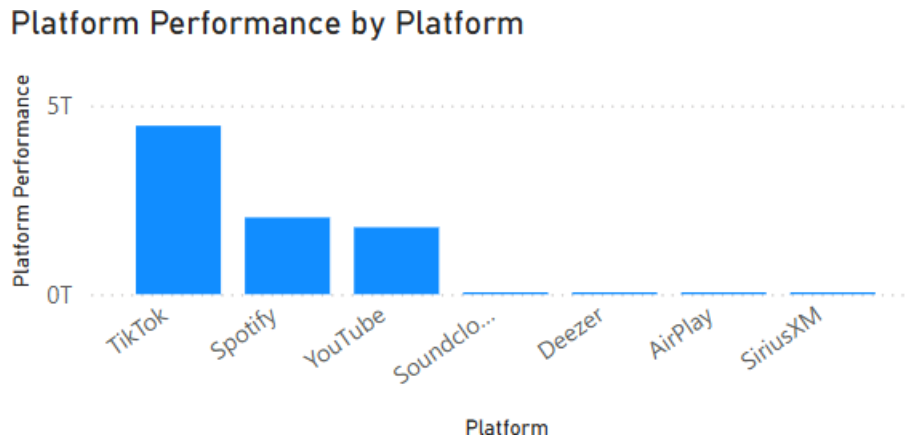
- Release year analysis

Total Songs by Release Year



2023 has the highest number of songs released in the dataset.
Total Songs - 1115 songs.

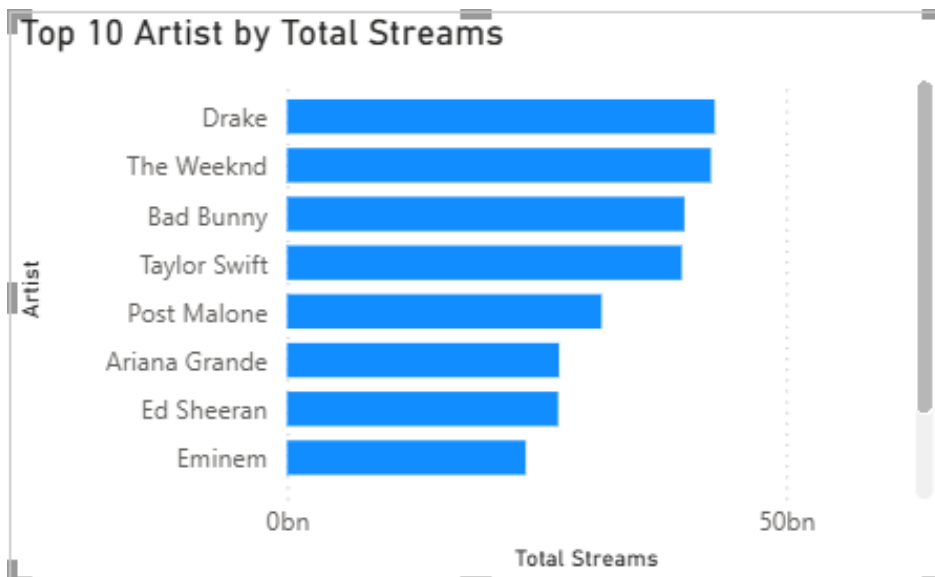
- Platform statistics



TikTok Platform contributes the most to song popularity.

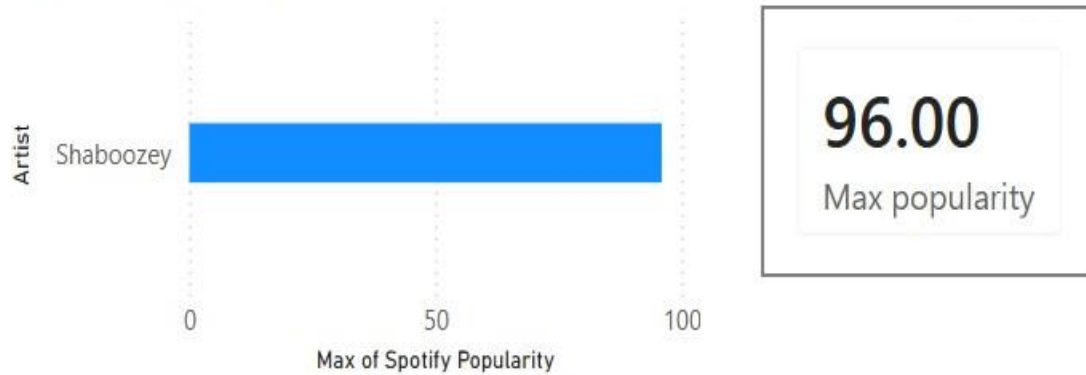
9. ARTIST PERFORMANCE ANALYSIS

- Top 10 artists by total streams



- **Highest popularity artists**

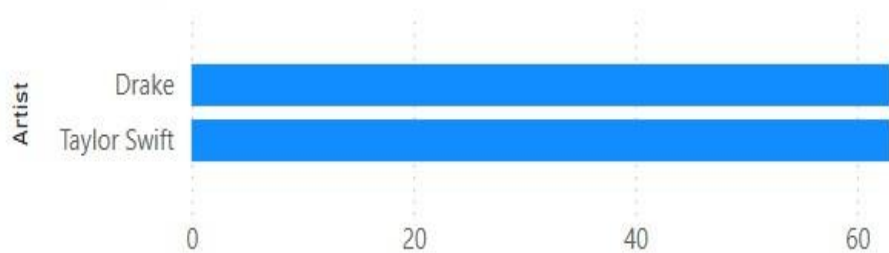
Highest Popularity by Artist



Shaboozey
Highest Popularity of 96

- **Most frequently appearing artists**

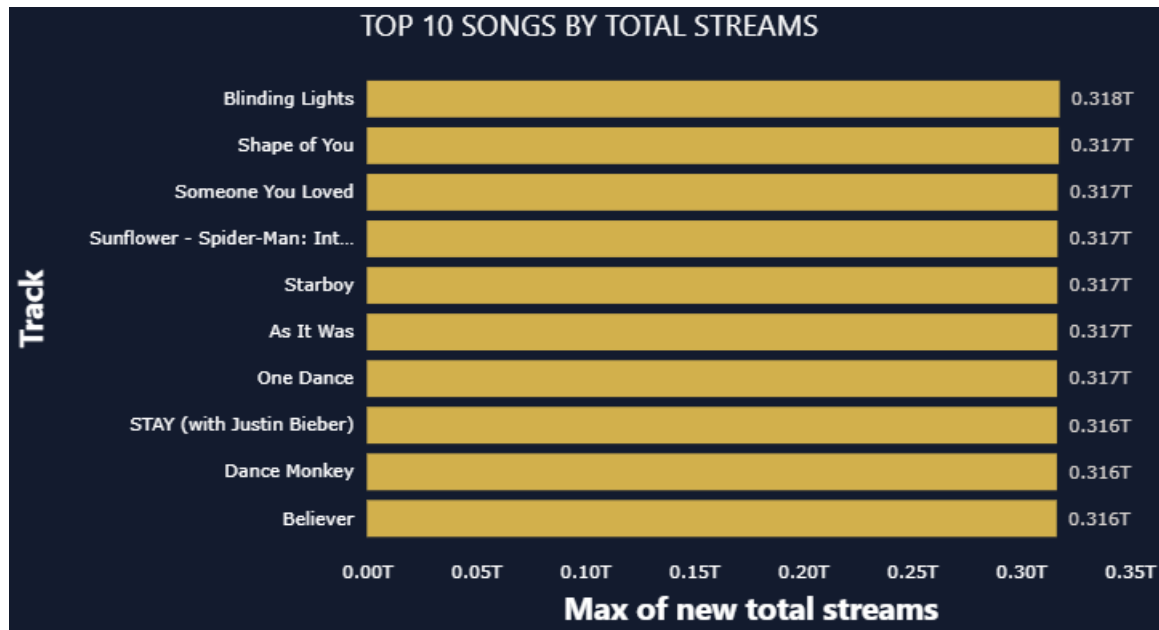
Most Frequently Appearing Artist



**Drake and Taylor Swift appear most frequently in the dataset,
with 63 songs in each.**

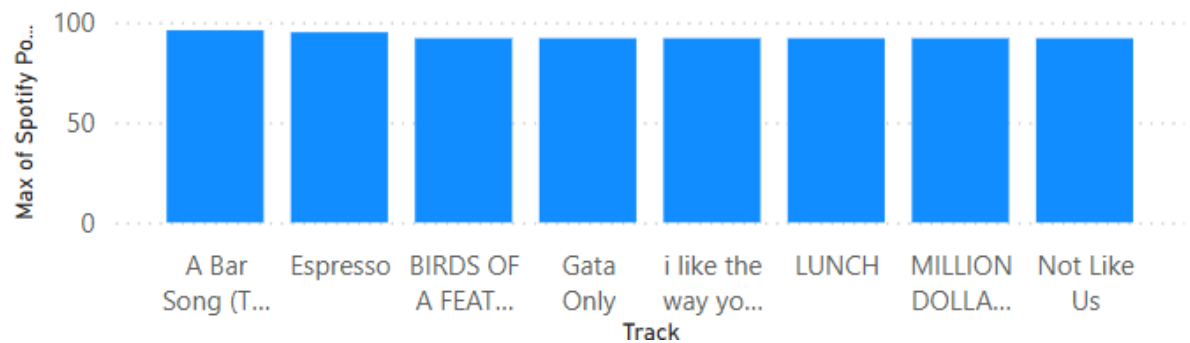
10. SONG PERFORMANCE ANALYSIS

- Top streamed songs

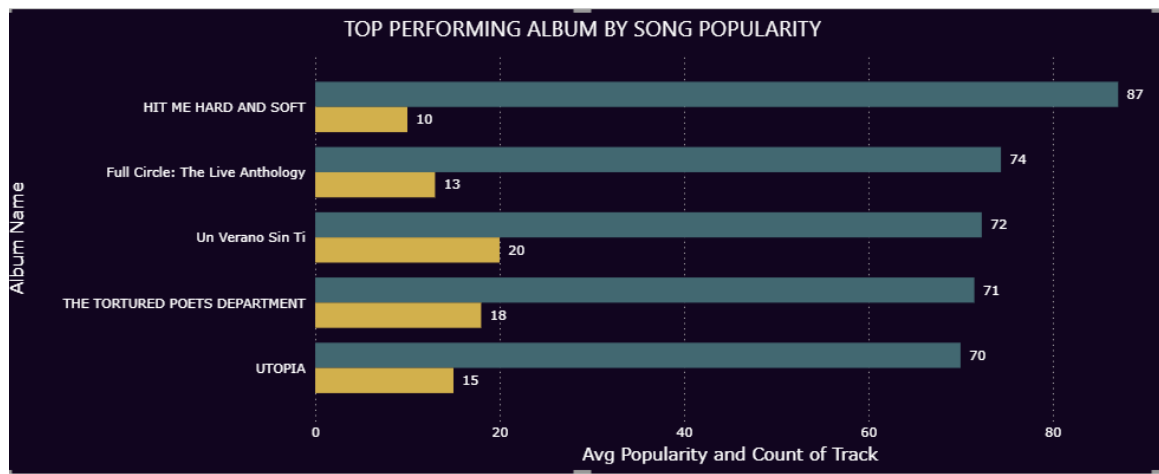


- Most popular songs

Highest Popularity of Songs



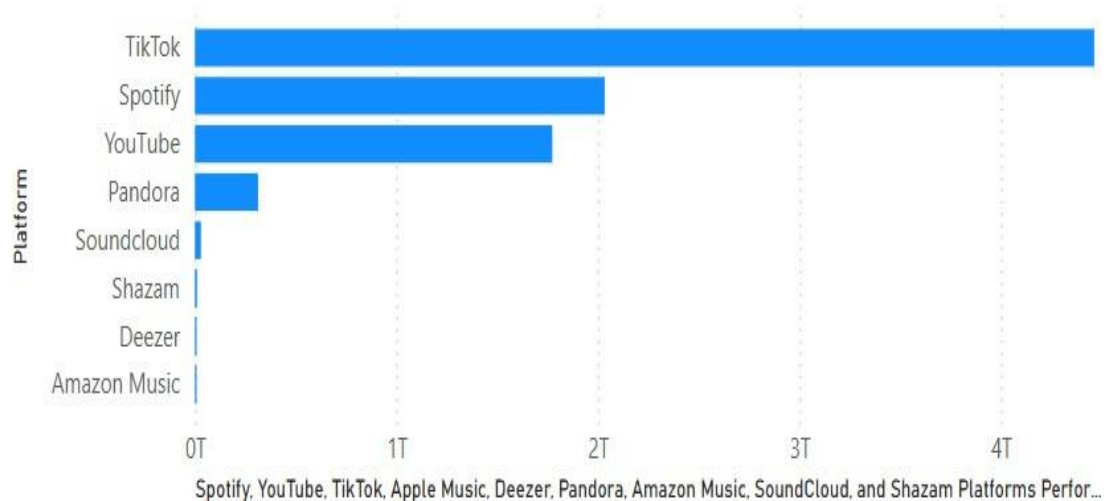
- Albums containing the highest-performing tracks



11. PLATFORM ANALYSIS

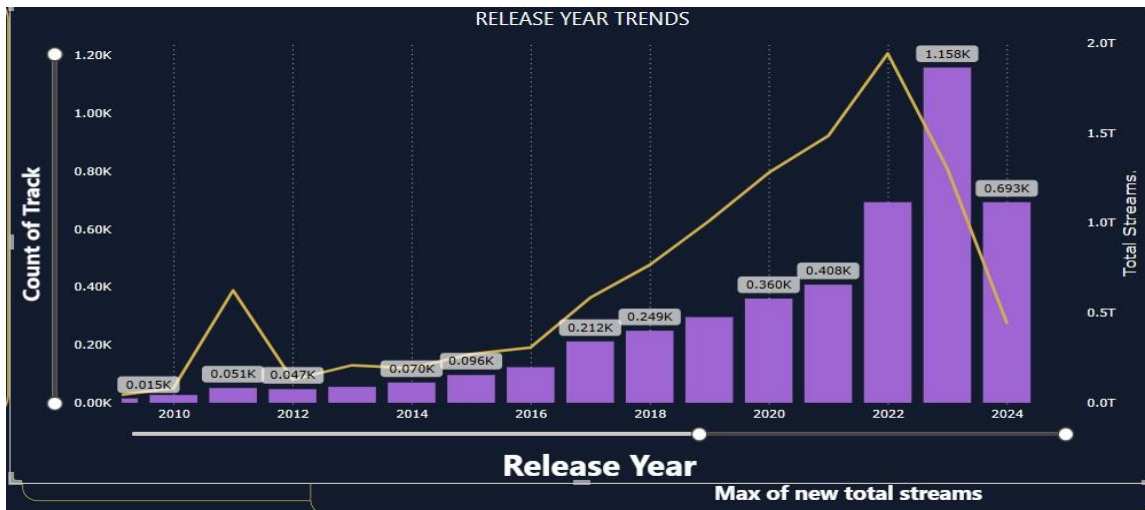
Compared Spotify, YouTube, TikTok, Apple Music, Deezer, Pandora, Amazon Music, SoundCloud, and Shazam.

Spotify, YouTube, TikTok, Apple Music, Deezer, Pandora, Amazon Music, SoundCloud, and Shazam Platforms Performance by Platform



12. TREND ANALYSIS

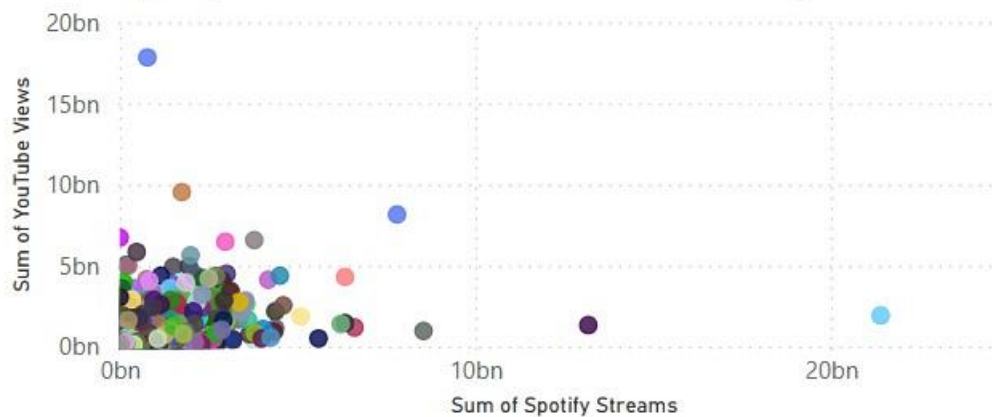
Analyzed release-year trends



13. RELATIONSHIP ANALYSIS

• Spotify Streams vs YouTube Views

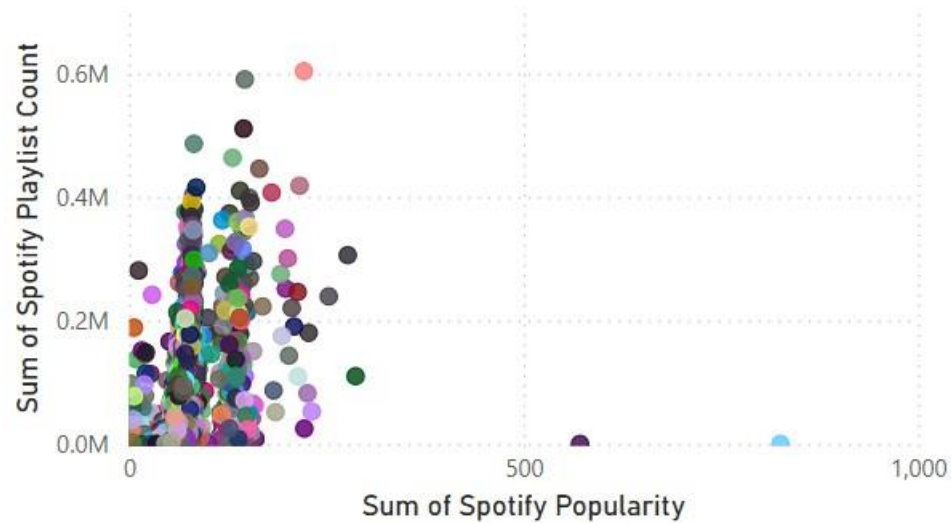
Sum of Spotify Streams and Sum of YouTube Views by Track



There is a moderate positive relationship between Spotify streams and YouTube views. Most songs with higher Spotify streams also have higher YouTube views.

- **Spotify Popularity vs Playlist Count**

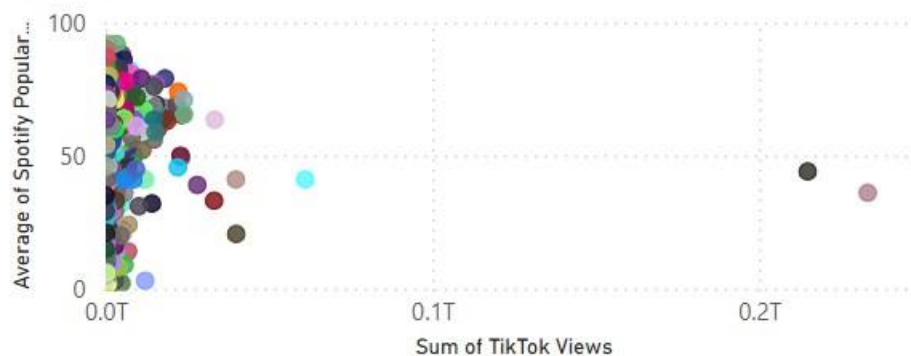
Sum of Spotify Popularity and Sum of Spotify Playlist Count by Track



Songs with higher playlist counts generally have higher Spotify popularity. This indicates that playlist placements can contribute to increasing a song's popularity.

- **TikTok Views vs Spotify Streams**

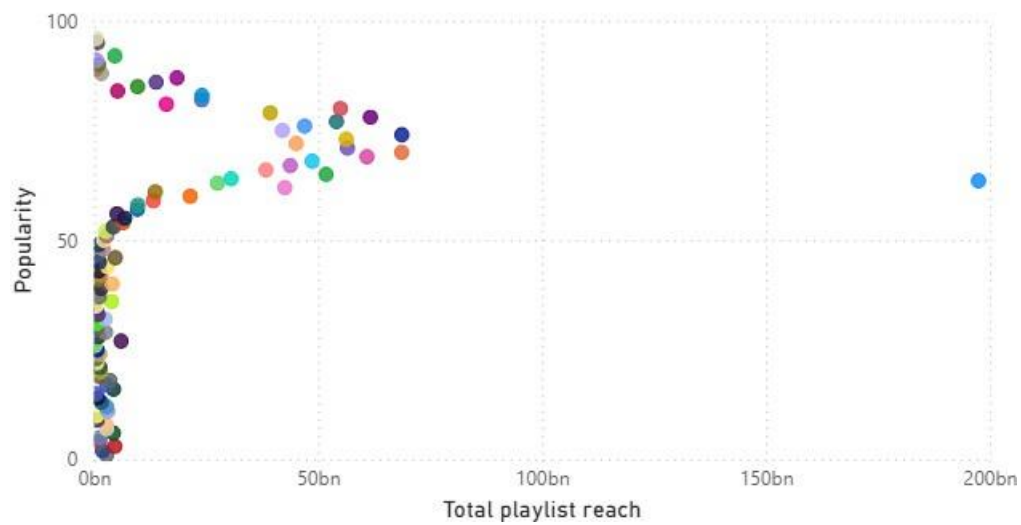
Sum of TikTok Views and Average of Spotify Popularity by Track



TikTok views have a moderate impact on Spotify popularity. Some songs with more TikTok views have higher Spotify popularity, while other.

- **Playlist Reach vs Popularity**

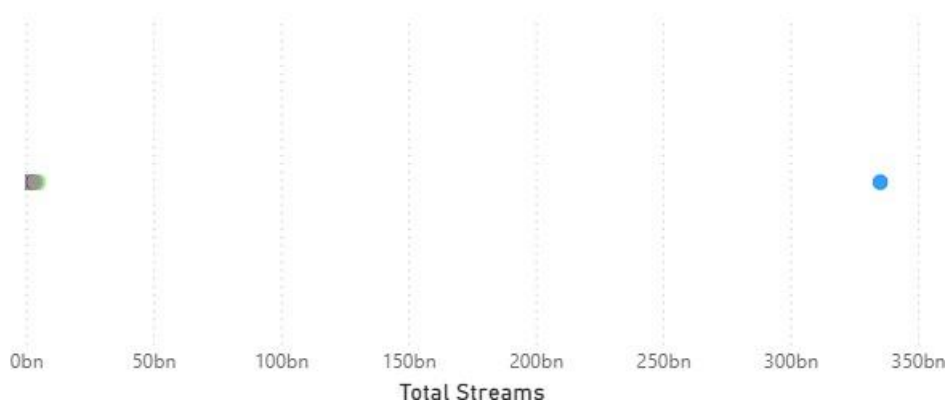
Total playlist reach by Popularity



Higher playlist reach generally leads to higher song popularity.

- **Shazam Counts vs Streaming Performance**

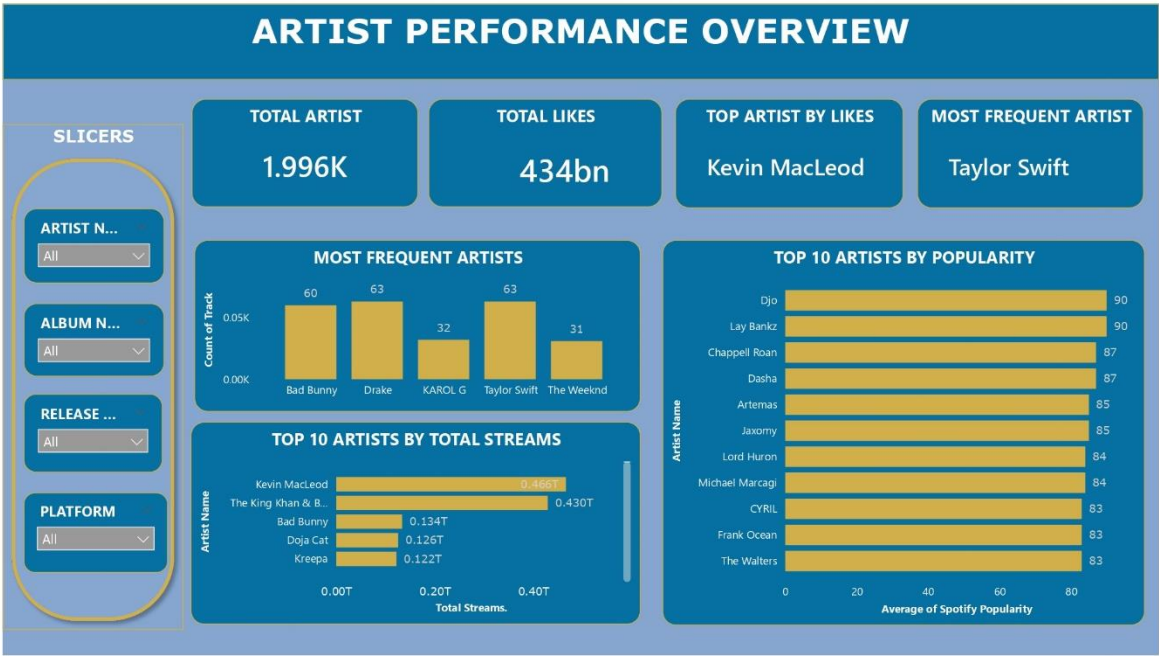
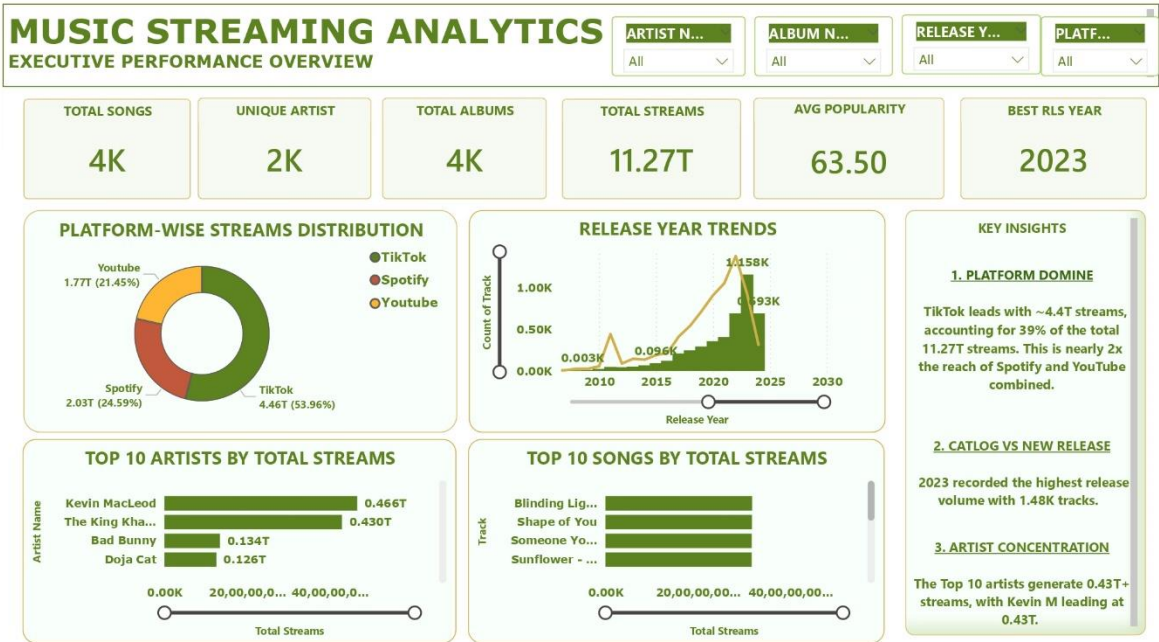
Total Streams by Shazam Counts

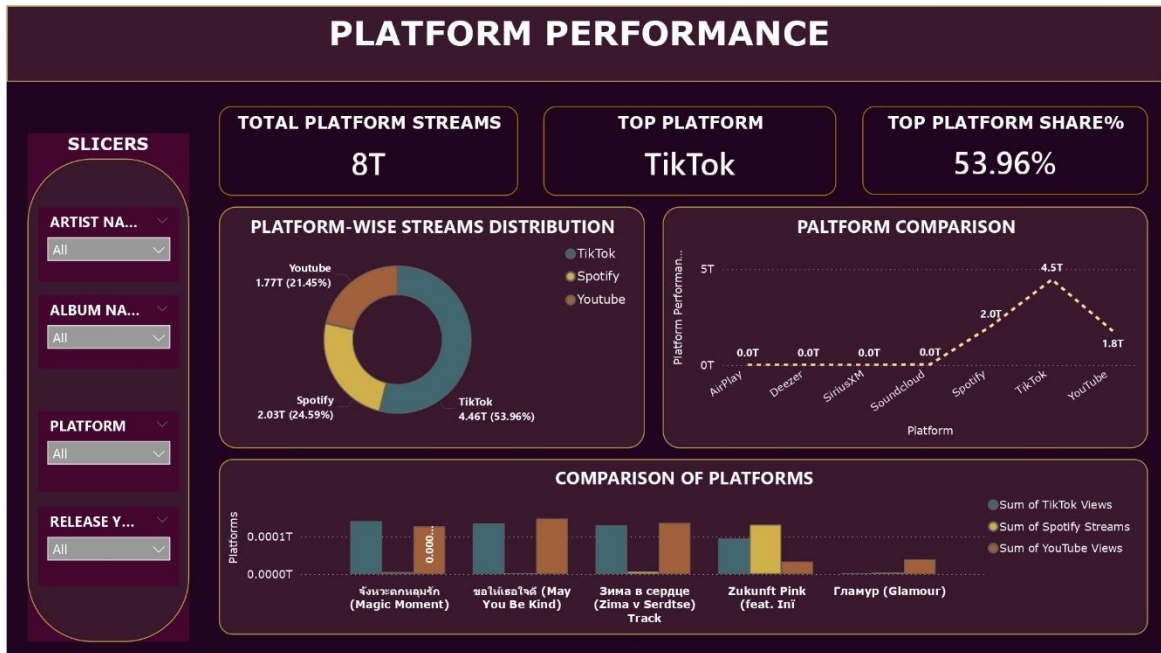
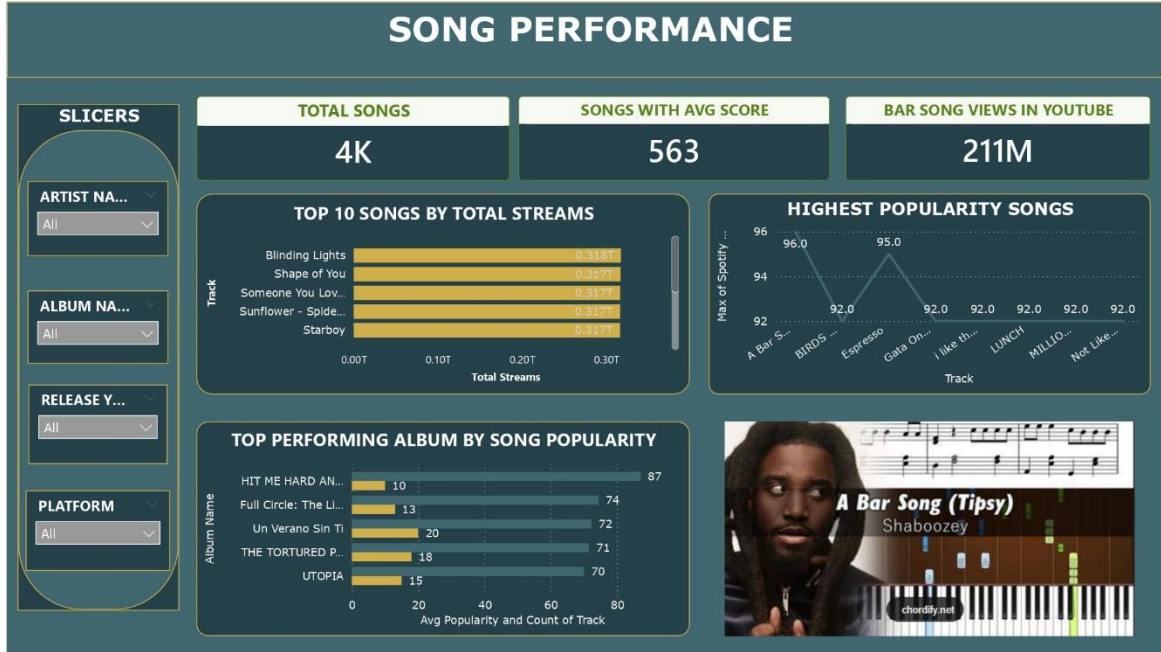


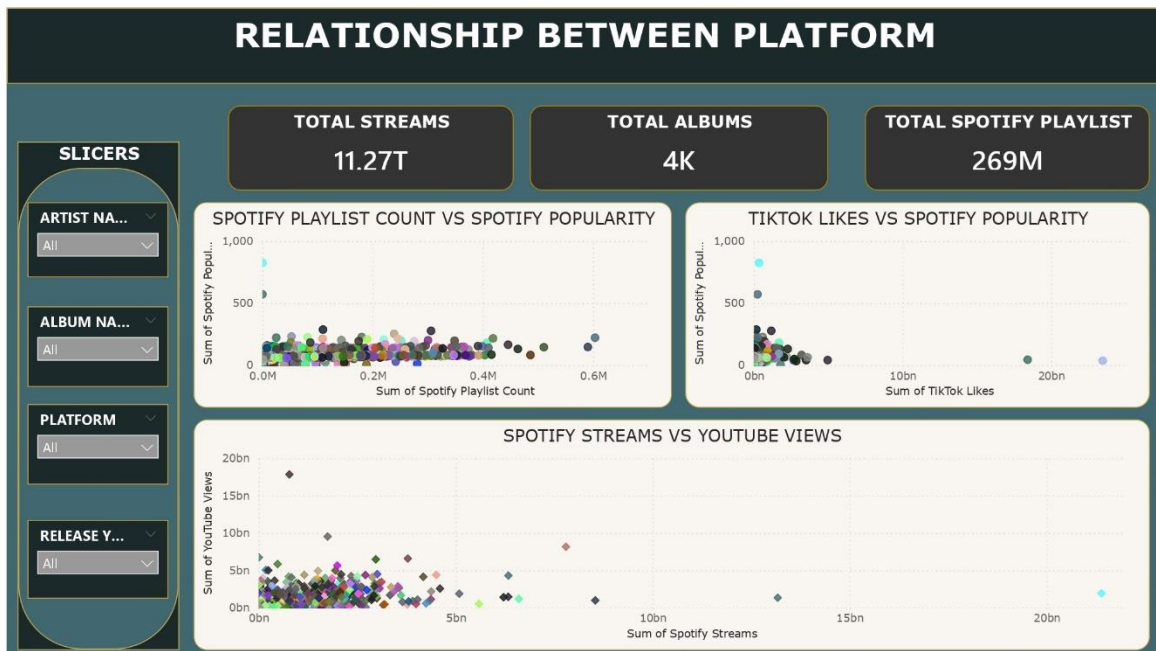
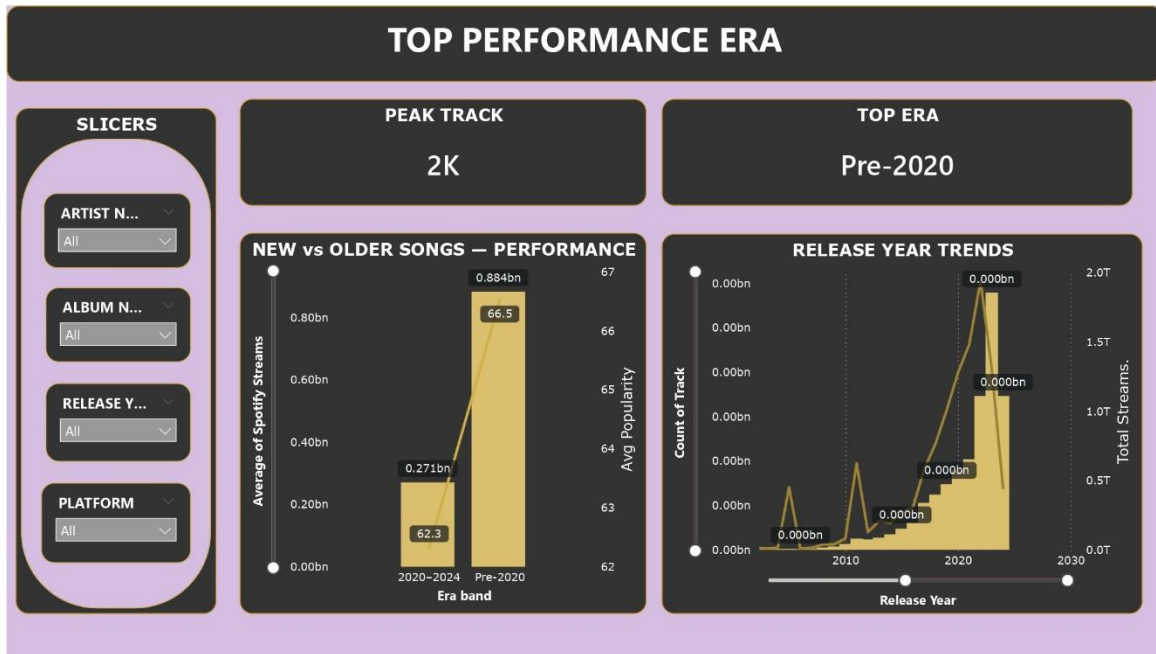
Higher streaming performance is associated with higher shazam counts.

Strong positive relationships were observed between playlist exposure and streaming success.

DASHBOARD PREVIEW







15. Business Insights



16. RECOMMENDATIONS

1. Investing in Spotify or YouTube to reduce platform risk
2. Monitor song performance
3. Release music strategically based on location
4. Shaboozey's "A Bar Song"(Tipsy) and collaborate with creators like Kevin MacLeod
5. Blinding Lights by Weekend
6. Comparison between Platforms by Artist

18. CONCLUSION

This project successfully transformed raw music streaming data into actionable business insights using Excel and Power BI. The dashboard supports data-driven decisions for artists, record labels, and streaming platforms.